

Low Cost Quantum Key Distribution (QKD) Simulator using Photonic Communication

Presented By
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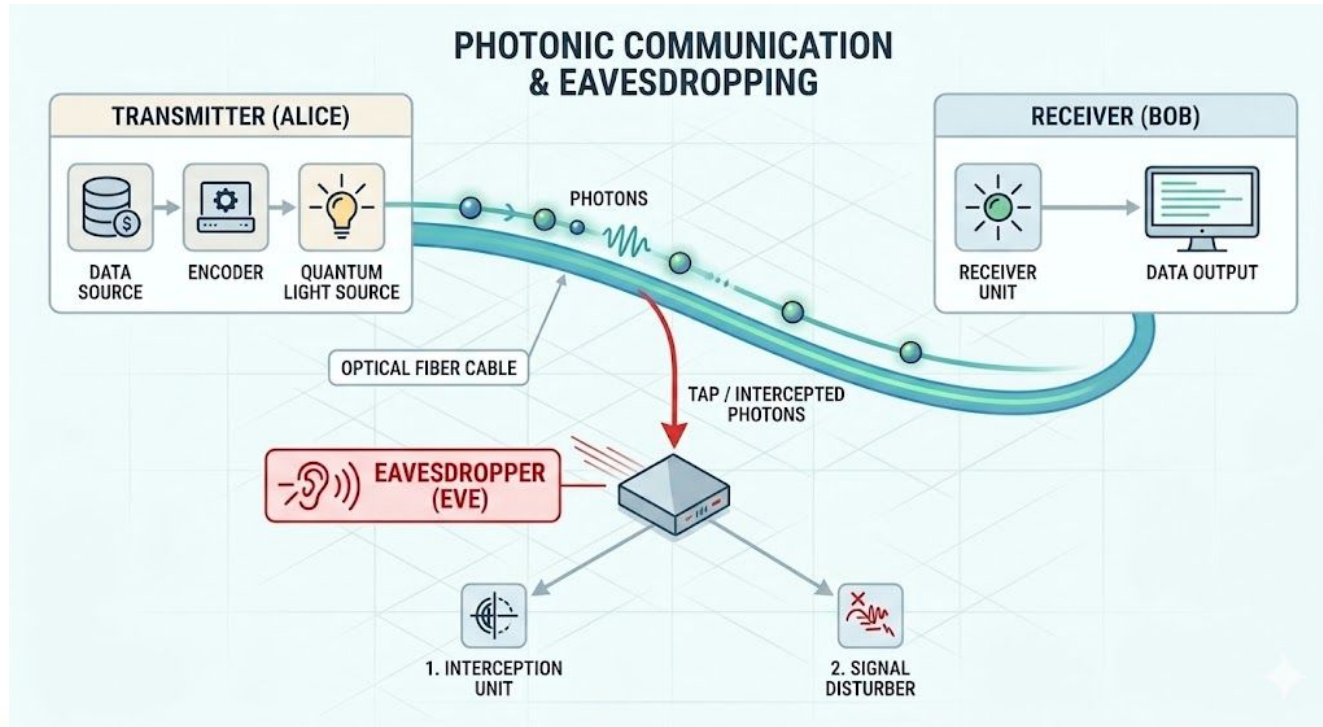
Why Quantum Key Distribution?

- Current Problem
 - Classical cryptography relies on computational hardness
 - Future quantum computers threaten RSA and ECC
 - Sensitive communications require long-term security
- QKD Solution
 - Uses principles of quantum mechanics
 - Eavesdropping changes quantum states
 - Enables secure key exchange with intrusion detection
- Real-World Applications
 - Banking and financial networks
 - Government communication
 - Defense communication systems
 - Critical infrastructure
 - Data center interconnects

Quantum Key Distribution (QKD)

- What is QKD?
 - QKD is a secure communication method that uses quantum states to distribute cryptographic keys.
- Core Principles
 - Photons carry information
 - Quantum measurement disturbs state
 - No-cloning theorem prevents copying
 - Eavesdropping becomes detectable

QKD over Photonic Communication



QKD over Photonic Communication

1. Photons Carry Information

- A photon is a tiny packet of light energy. In quantum communication, photons are used as the “messengers” that carry information from Alice to Bob
- In normal digital communication:
 - bits are represented using voltage levels, such as 0 V and 5 V.
- In quantum photonic communication:
 - bits are represented using quantum properties of photons,
 - especially polarization
- Polarization as Information
 - A photon’s electric field can oscillate in different directions.
 - For example:

Polarization	Meaning
Horizontal	Bit 0
Vertical	Bit 1
45°	Bit 0 in diagonal basis
135°	Bit 1 in diagonal basis
 - Thus, information is encoded into photon states.

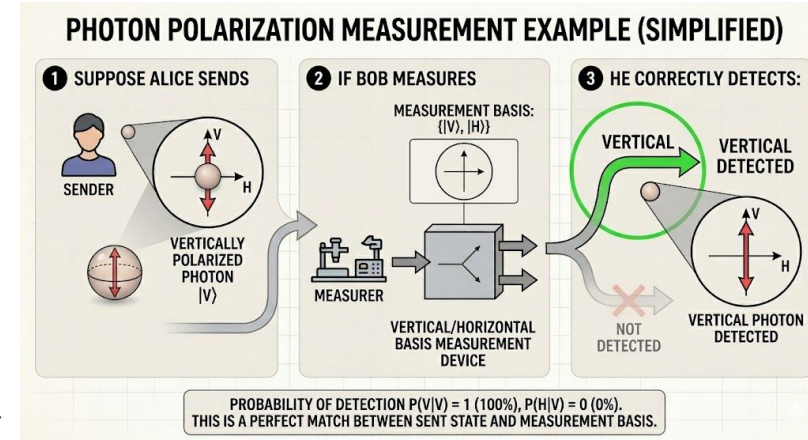
QKD over Photonic Communication

2. Quantum Measurement Disturbs the State

- This is one of the most important ideas in quantum mechanics
- In classical systems: you can measure something without changing it
- But in quantum systems: measurement itself changes the quantum state
- Example with Polarization

Suppose Alice sends: a vertically polarized photon

If Bob measures using: vertical/horizontal basis, he correctly detects: Vertical



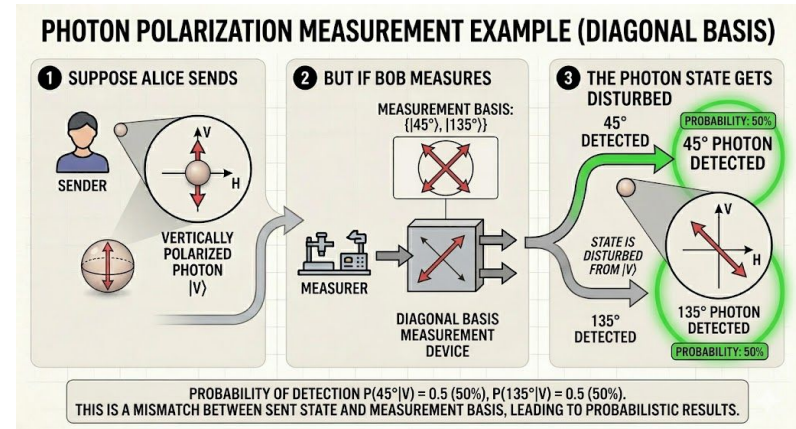
QKD over Photonic Communication

2. Quantum Measurement Disturbs the State

- This is one of the most important ideas in quantum mechanics.
- In classical systems: you can measure something without changing it.
- But in quantum systems: measurement itself changes the quantum state
- Example with Polarization

Suppose Alice sends: a vertically polarized photon

*But if Bob measures using: diagonal basis, the photon state gets disturbed.
Now the result becomes probabilistic: sometimes 45° , sometimes 135°*



QKD over Photonic Communication

2. Quantum Measurement Disturbs the State

- Why This Happens
 - Quantum particles do not always have fixed classical properties before measurement
 - Measurement “forces” the system into one of the allowed outcomes
 - This process is called: Wave function collapse
- Importance in QKD
 - This property creates security. If Eve tries to measure photons: she may choose wrong basis, measurement disturbs the photon, errors appear in Bob’s received bits. Thus: spying changes the communication

QKD over Photonic Communication

3. No-Cloning Theorem Prevents Copying

- In classical communication: data can be copied perfectly. For Example: copying a file, duplicating packets, forwarding emails.
- But in quantum mechanics:
 - an unknown quantum state cannot be copied perfectly.
This is called: No-Cloning Theorem

QKD over Photonic Communication

3. No-Cloning Theorem Prevents Copying

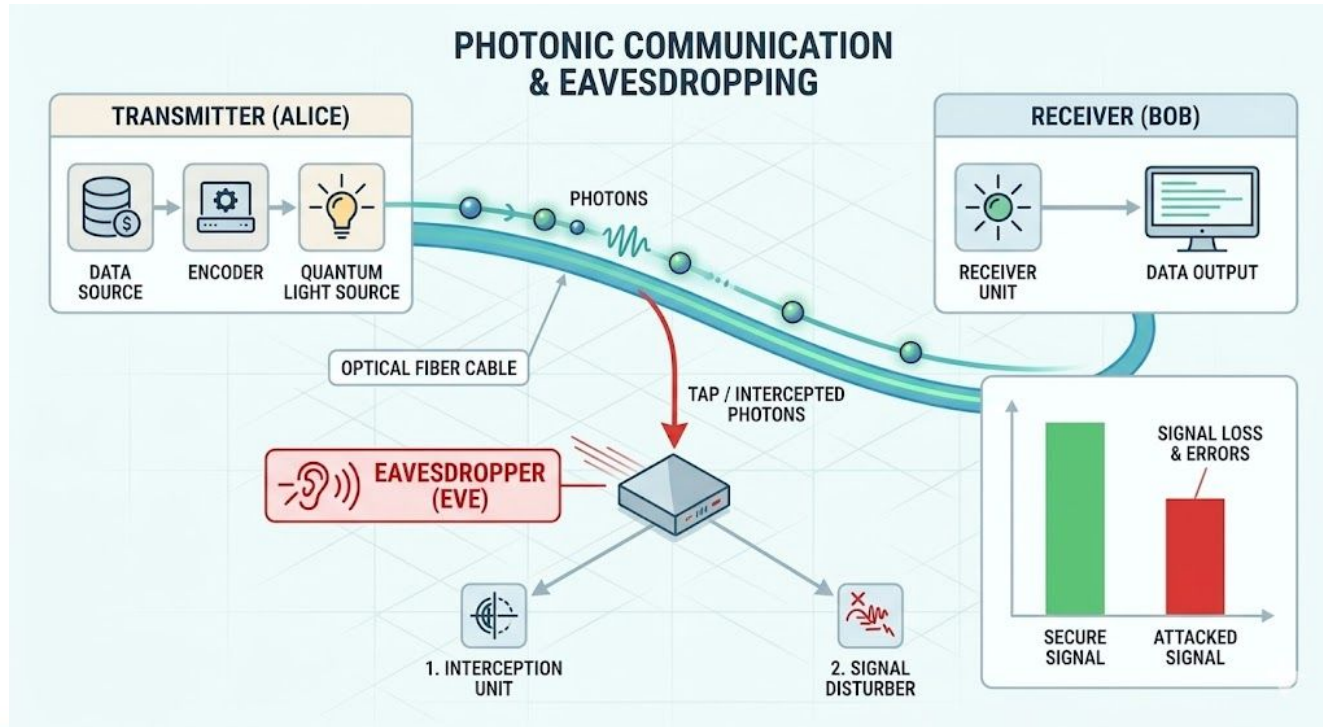
- Intuition
 - Suppose Eve intercepts a photon. She wants to: keep a copy and send another identical copy to Bob.
 - Quantum mechanics does not allow perfect duplication of an unknown quantum state.
 - Why?
 - A quantum state may exist in:
 - superposition, unknown polarization, probabilistic form.
 - And Copying would require: measuring it exactly first. But measurement itself disturbs the state. So perfect copying becomes impossible.

QKD over Photonic Communication

3. No-Cloning Theorem Prevents Copying

- Security Benefit
 - This is a major reason QKD is secure.
 - Eve cannot: secretly duplicate photons, store copies, and later analyze them.
 - She must interact with the original photon, which introduces detectable errors..

QKD over Photonic Communication



QKD over Photonic Communication

Spying leaves evidence.

- **How Eve Causes Errors:** Suppose **Alice** sends Vertical polarization and **Eve** intercepts it. But Eve does not know the correct basis
 - Case 1: Eve Chooses Correct Basis: She measures correctly. No error introduced.
 - Case 2: Eve Chooses Wrong Basis: Measurement disturbs photon state.
 - When Bob later measures: he may get wrong result.
 - Thus: error rate increases.
- **Quantum Bit Error Rate (QBER):** Alice and Bob compare a small subset of bits publicly. If too many mismatches occur: they conclude: Possible eavesdropping detected
- **Example:**

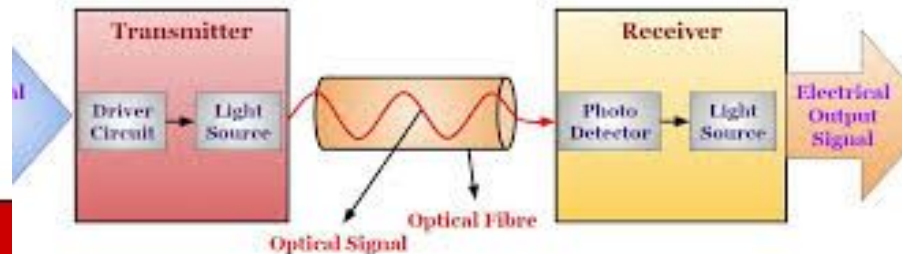
	Transmission	Error Rate
Without Eve:	Normal	Very low
With Eve:	Intercept-resend attack	High
- **Why is this Revolutionary?**
 - Traditional security depends on: computational difficulty
 - Quantum security depends on: laws of physics. That makes QKD fundamentally different from classical cryptography

Basics of Photonic Communication

Photonic/Optical Communication

- Components:
 - Transmitter: LED or Laser source
 - Channel: Optical fiber or free-space optical link
 - Receiver: Photodiode (PIN/APD) converts light to electrical signals
 - Signal processing and decoding

Principle of Optical Fibre Communication



Signal Processing in Photonic Communication?



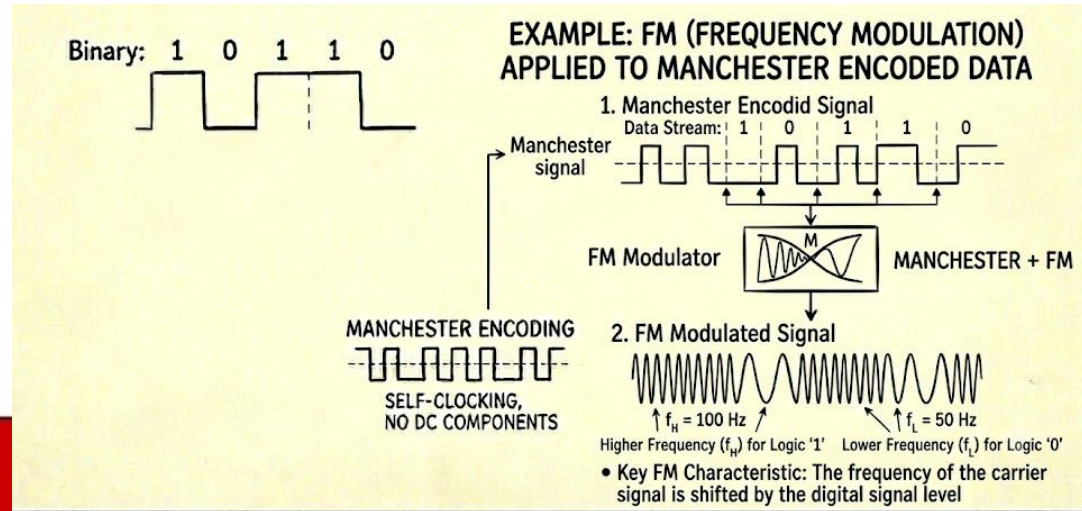
Think if someone is far away. What do you do? You use modulation and coding.
What type of modulation is used in this figure?

Photonic/Optical Communication

- How Information is carried:
 - Digital encoding to increase data rate.
 - Analogue modulation to avoid fading

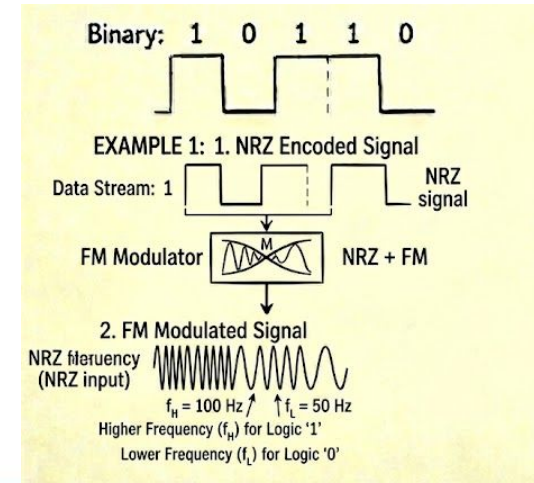
Why Coding in Communication?

- Coding Schemes allows more data embedding in each signal element. Same as use of “more syllables”.
- Schemes:
 - NRZ
 - RZ
 - Manchester ...



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Coding Schemes in Photonics

Scheme	Main Advantage	Main Disadvantage
NRZ	Minimum bandwidth	Poor synchronization
RZ	Better synchronization	Higher bandwidth required
Manchester	Self-clocking	Highest bandwidth required

Why Modulation in Communication?

- Allows smaller antennas
- Enables long-distance transmission
- Prevents signal mixing (interference)
- Improves signal quality and reduces noise
- Supports multiplexing
- Efficient use of bandwidth

$$L \approx (\lambda/4) = (c/4f)$$

where: L = antenna length; λ = wavelength; c = speed of light; f = frequency

Why Modulation in Communication?

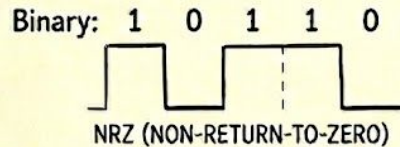
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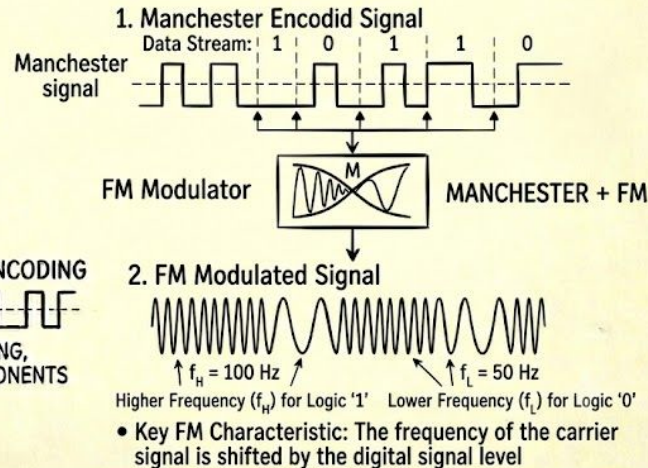
Think if someone is far away. What do you do? You use modulation.
What type of modulation is used in this figure?

Modulation and Coding Scheme (MCS)

LINE CODING AND MODULATION METHODS IN OPTICAL COMMUNICATION

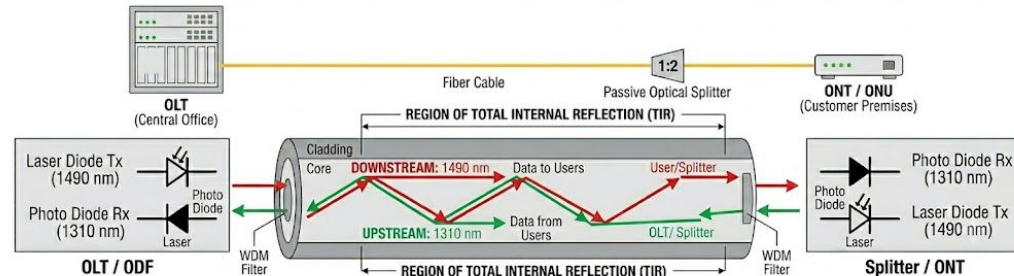


EXAMPLE: FM (FREQUENCY MODULATION) APPLIED TO MANCHESTER ENCODED DATA



Photonic/Optical Communication

- Modulation: Intensity, Phase, Frequency
- Digital data represented by optical pulses
- Modulation:
 - Wavelength Division Multiplexing (WDM) $\Rightarrow$ FDM
 - Used in GPON Gateways for Duplex communication
 - TX: 1490 nm; Rx: 1310 nm



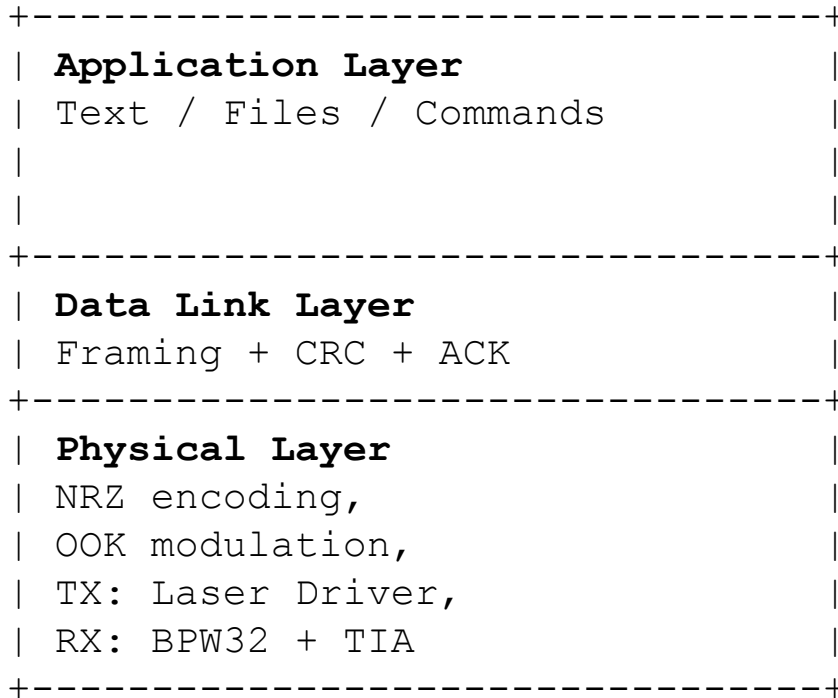
Basics of TCP/IP Stack

TCP/IP Protocol Stack

- The TCP/IP protocol stack (also known as the Internet Protocol Suite) is the conceptual framework and set of rules that allows computers to communicate over a network, including the global Internet.
- It breaks the complex process of moving data across the world into a modular, 4-layer stack. As data moves down the stack on the sender's side, each layer adds its own formatting header (encapsulation). When it arrives at the receiver, the process reverses (decapsulation).

+-----+		
	Application	
	HTTP, MQTT, DNS	
+-----+		
	Transport	
	TCP, UDP	
+-----+		
	Internet	
	IP, ICMP	
+-----+		
	Data Link	
	802.11 MAC, Ethernet	
+-----+		
	Physical	
	RF, Fiber, Copper	
+-----+		

TCP/IP Protocol Stack



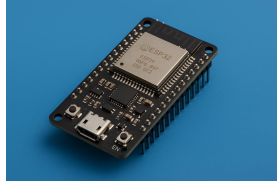
Development of full stack will be difficult.

Therefore, we'll create a smaller stack like this.

Components required:

Our Goal

Our Setup: Hardware Requirement



Components required: for TX

ESP32 DevKit → Controller

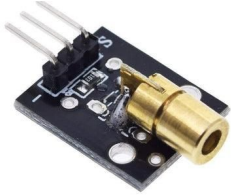
Laser Diode Module → Optical transmitter

NPN Transistor (2N2222/BC547) → Laser driver

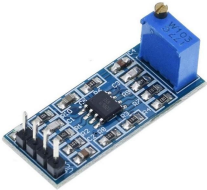
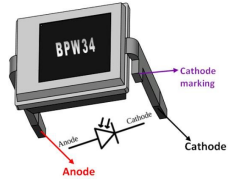
220Ω–1kΩ Resistor → Transistor bias

Breadboard/PCB → Prototyping

Power Supply (5V) → Laser power



Our Setup: Hardware Requirement



Components required: for RX

ESP32 DevKit → Controller

Breadboard/PCB → Prototyping

Power Supply (5V) → Laser power

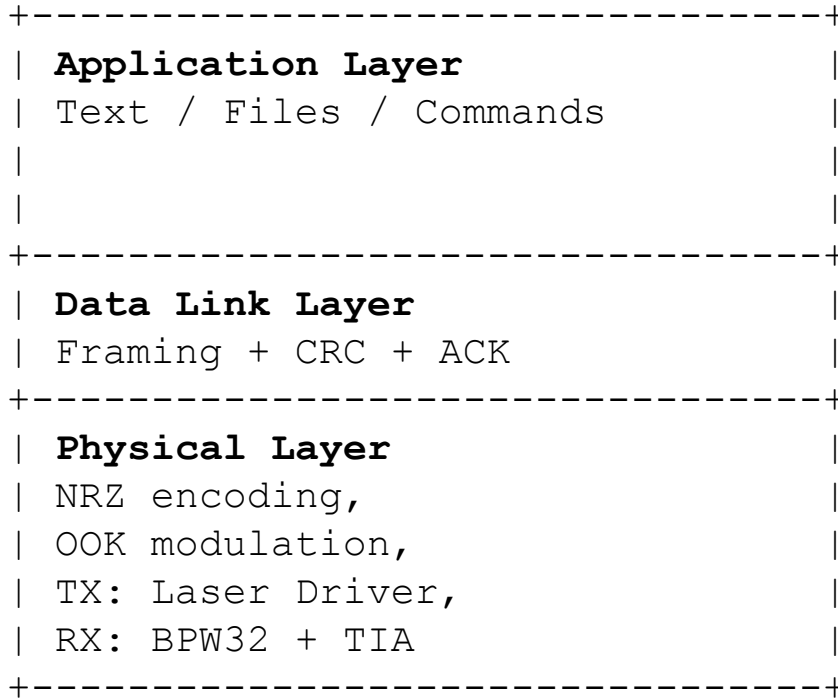
BPW32 Photodiode → Optical detector

LM393 Comparator → Digital slicing

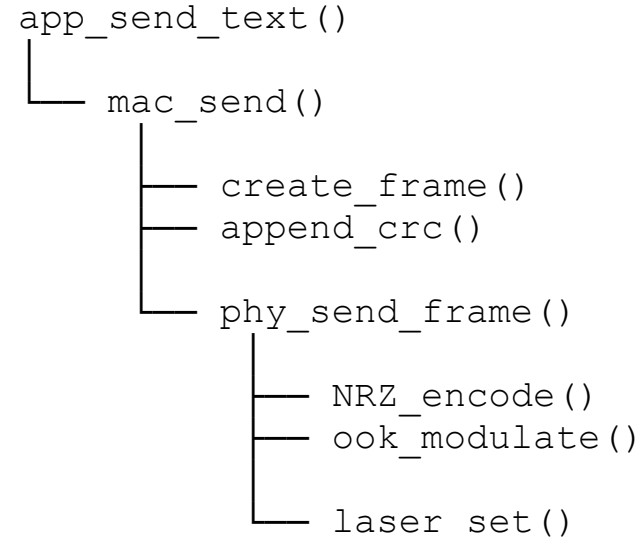
Feedback Resistors ($100\text{k}\Omega$ – $1\text{M}\Omega$) → Gain setting

Capacitors (10pF – 100nF) → Stability/filtering

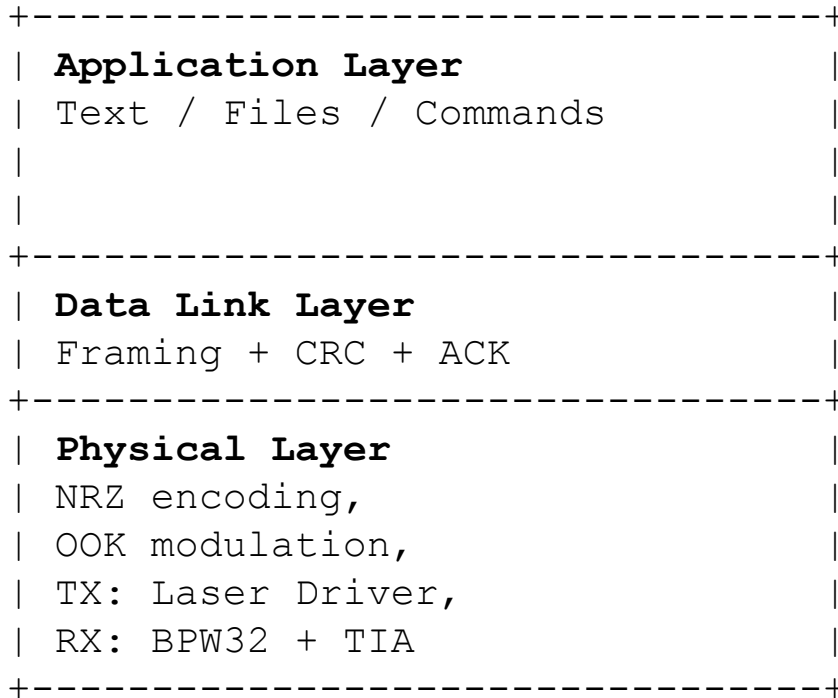
Our Setup: Software Requirement



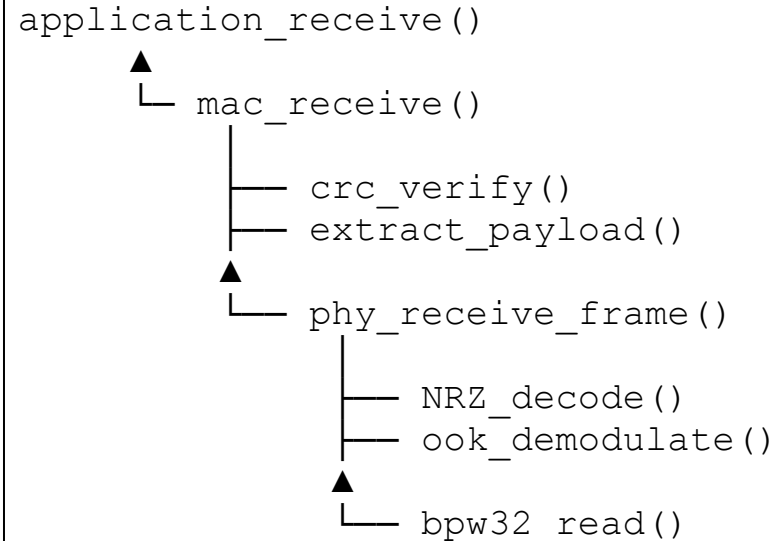
Complete TX Call Flow



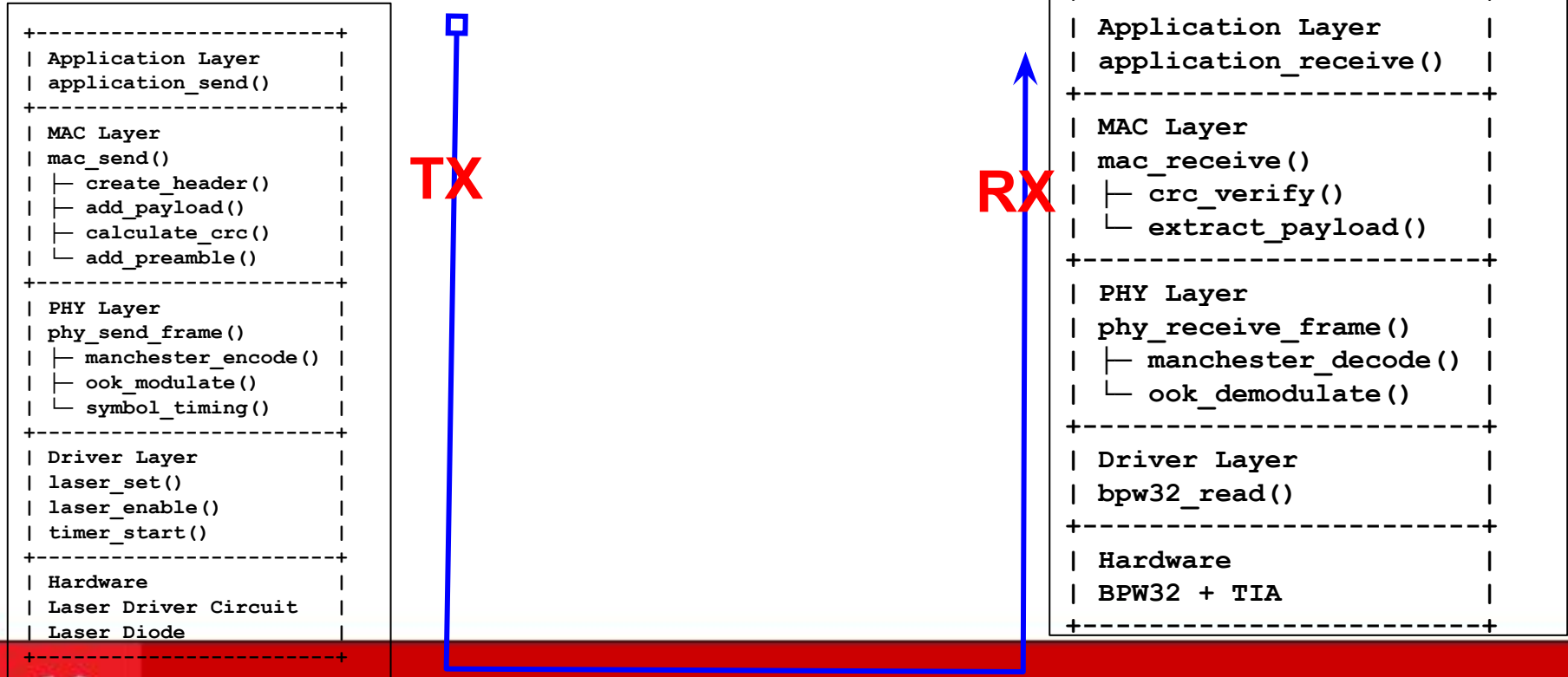
TCP/IP Protocol Stack



Complete RX Call Flow



TCP/IP Protocol Stack



Important Tutorials

Tutorials

- Lec05: Digital Communication for Optical Communication
 - <https://www.youtube.com/watch?v=j0c9xHWtATo>
- Arduino Photodiode Light Sensor (BPW34) Circuit & Code Example. Arduino Photodiode Circuit Example For light intensity measurement (BPW34)"
 - <https://deepbluembedded.com/arduino-photodiode-light-sensor-bpw34-circuit-code-example/>
- Li - fi communication system using ON OFF Keying (OOK)
 - <https://www.youtube.com/watch?v=Aj9jUR7Xog8>
- 28BYJ-48 Stepper Motor with ULN2003 & Arduino | Full Step vs Half Step
 - <https://www.youtube.com/watch?v=uJriwdnPUD4>
- Quantum cryptography: The BB 84 protocol (U3-02-03)
 - <https://www.youtube.com/watch?v=2kdRuqvlaww>